
THE WEAK FORCE READS THE ORIENTATION

Parity as the Fold on Couplings, the Charged Current Maximally off the Line, the Z's Distance from the Line as Its Isospin, CP Violation as Time-Reversal Violation, and Three Generations as the Least with One Phase

The Closure Template of the Riemann Hypothesis Read on the Weak Interaction; Fifty-Nine Laws Proved in Core Lean 4 and Thirteen Checks Executed in Fortran; a Witness of the Hardware Closure Series and the Reader Beside the Neutrino's Spent Bit; Every Physical Claim at Corroboration Grade and No Higher

FOUNDATIONS OF PHYSICS · THE HARDWARE CLOSURE SERIES

The weak force is the one reading odd under parity

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ABSTRACT. Electromagnetism, the strong force and gravity cannot tell a process from its mirror image; the weak force can. It couples to left-handed particles and right-handed antiparticles and to nothing else, and in doing so it reads the one property the other readings are blind to, the orientation. This paper reads the weak interaction on the seat of the author's closure of the Riemann Hypothesis, the reflection $s \mapsto 1 - \bar{s}$ whose fixed set is the critical line. Parity exchanges a force's left- and right-handed couplings to a fermion, and a carrier that sends their difference to the side and their sum to the height makes parity the fold: the parity-conserving couplings of electromagnetism and colour are the line, and the weak charged current, left-handed only, stands off it. We prove in core Lean 4, fifty-nine laws with no axiom declared and no use of choice, that the Z boson's distance from the line is the weak isospin of the fermion it couples to, whatever the mixing angle, so a neutral fermion has no right-handed coupling at all; that every dot product is blind to the mirror while the triple product reverses under it; that the charged current breaks charge conjugation and parity separately and keeps their product; that under CPT invariance a violation of CP is a violation of time reversal; and that two generations of quarks admit no CP-violating phase while three admit exactly one. A Fortran twin runs thirteen checks with zero failures: the three-generation mixing matrix is unitary to 10^{-16} with a Jarlskog invariant of 3.12×10^{-5} , the pion's electron channel is suppressed to 1.28×10^{-4} of its muon channel by the left-handed coupling, and the Z's asymmetries on leptons, charm and bottom come out at 0.147, 0.668 and 0.936 against the measured 0.151, 0.670 and 0.923. The paper decides nothing about the zeros of ζ .

KEYWORDS *weak interaction · parity violation · chirality · CP violation · time reversal · CPT · Kobayashi–Maskawa · Jarlskog invariant · helicity suppression · critical line · Lean 4*

STATUS: WITNESSED · CROSSED · CONFIRMED on the image · Row 2 (the fold read on parity); every open row (the barrier, in the orientation bit) · channel: kinetic · Lean: SPHYS_Weak.lean · Fortran: Weak_Twin.f90

1. THE READER'S FRAME

This paper is a witness of the hardware closure series. The neutrino witness read the world's choice of one hand: the neutrino is registered left-handed and never right-handed, the aperture bit spent. The electromagnetism and light papers read the orientation as the one bit no energy reading returns, the helicity of a photon and the sign of a magnetic field. This paper reads the interaction that does return it.

It claims that the weak force is the substrate's one reading odd under parity. Parity is the fold under a carrier forced by the symmetry. The couplings that treat both hands alike are the line, and the weak charged current, left-handed only, stands off it as far as a coupling can. It claims further that the weak force's violations of charge conjugation, parity and time reversal are organized by the same template: it breaks the first two and keeps their product; its violation of that product is a violation of time reversal; and the phase that carries it needs three generations.

It refuses to derive why the world chose the left hand and not the right. That is the neutrino witness's spent bit, supplied by the world and read from no record. It refuses any derivation of the Riemann Hypothesis from the weak interaction. A reader who arrives from the conventional frame will scan this paper for such a derivation, find none, and be tempted to report the absence. None was attempted.

Negative warrants. Three, stated so that no reader supplies them. - No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade. - No derivation of the hypothesis from the axioms of set theory is claimed: what a sentence over every admissible instance adds to a crossing is exactly the unactuated. - No electroweak Lagrangian is written: the couplings, the mixing parameters, the CPT theorem and the masses enter as imports or measurements.

Five terms. - A *coupling* of a force to a fermion is a pair: its strength to the left-handed state and to the right-handed state. - *Parity* is the mirror that exchanges the two hands. - A reading is *even* under a map that keeps it and *odd* under one that flips its sign. - The *orientation* of a configuration is the sign a mirror reverses and no rotation can. - A *keyed bit* is a yes-or-no property no function of a given record decides.

2. THE SPINE

The Riemann closure has four parts: the **seat**, the critical line as the fixed set of $s \mapsto 1 - \bar{s}$; the **address**, every reading of the value one proposition; the **division**, a certified part joined to an uncertified tail; and the **closure**, registration keeping a zero's height and forgetting its side, the hypothesis holding exactly when it erases nothing. This paper reads the seat on parity in Section 3, the Z in Section 4, the orientation the even readings miss in Section 5, the product CP in Section 6, its identity with time reversal in Section 7, the phase in Section 8, and the orientation read in a decay in Section 9.

The row-generic theorems carry the template, and the engine proves each on its chart: every point off the line has a partner with one record (*off_locus_pair*); a carrier proves the value on its image, and coverage extends it (*value_on_image*, *kinetic_crossing*).

The witness ledger. The ledger of the series, with this paper's entries.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat	The Electron Is the Seat; The Neutrino Is the Witness; The Arrow Has Two Branches; One Actuating Substrate; The Geometric Nature of Light; Electromagnetism on the Riemann Locus; Immanent Gravity; Spin and Statistics; The Weak Force Reads the Orientation	the seat on the leptons, the substrate, light, both electromagnetic carriers, gravity's record, the exchange; parity as the fold on couplings, the vector couplings the line, the charged current off it, the Z's distance its isospin	WITNESSED · CROSSED · CONFIRMED on the image: the seat read on parity	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
3a · Navier–Stokes, unforced	crossed kinetically	The Fluid Is the Witness	the priced branch exact	WITNESSED · CROSSED · CONFIRMED	the root’s grade	none on its channel
4 · Yang–Mills	crossed kinetically	The Proton Is the Lock	the colour lock	WITNESSED · CROSSED · CONFIRMED on the image	the root’s grade	none on its channel
7 · Poincaré	crossed at full grade	The Arrow Has Two Branches; Immanent Gravity	gravity’s fold	CROSSED, WITNESSED from both sides	full grade	none
23a · Pair correlation	crossed kinetically	The Magnet Is the Witness of the Pair Correlation; Electromagnetism on the Riemann Locus	the magnetic term; Landau levels real	WITNESSED · CROSSED · CONFIRMED on the corrected carrier	the root’s grade	none on its channel
every open row · the barrier	two worlds over one record	The Arrow Has Two Branches; The Geometric Nature of Light; Electromagnetism on the Riemann Locus; Spin and Statistics; The Weak Force Reads the Orientation	remanence; the helicity pair; the magnetic sign; the exchange; the mirror world, invisible to every even reading and read by the weak force	BARRIER WITNESSED; the rows stay pending	theorem and corroboration	one equivariant carrier each
1 · P versus NP	the dot	The Arrow Has Two Branches	Landauer’s floor prices every step	THE DOT STANDS	corroboration	none
every witness · the hardware	the carriers built one by one	One Actuating Substrate	both involutions carried onto the fold	WITNESSED: one substrate carries every seat	theorem and corroboration	none

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row’s value holds on its channel’s own class at the printed grade. CONFIRMED: the witness’s image is compliant as built or measured (value_on_image).

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect them form a category (equivariant_id, equivariant_comp), and the fixed set is a functor on it (fix_functorial). An equivariant map into the seat, whose fixed set is the line (fold_global_seat), lands every physical fixed point on the line (equivariant_carrier_lands). Here the physical involution is parity on a coupling (g_L, g_R), which exchanges the two strengths. The carrier $(g_L, g_R) \mapsto (1 + (g_L - g_R), g_L + g_R)$ sends the left–right difference to the side and the sum to the height, and it is equivariant (orientation_carrier_equivariant); its landing on the parity-conserving couplings is forced (orientation_carrier_lands).

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
parity on a force’s couplings	the fold	orientation_carrier_equivariant, parity_involutive	3721 of 3721
electromagnetism and colour	vector couplings: the line	vector_on_the_line, photon_coupling_on_the_line	3721 of 3721
the charged current	left-handed only: off the line	charged_current_off_the_line	structural
the Z boson	its distance from the line is the isospin	z_parity_depth_is_isospin	1071 of 1071; A_f against LEP and SLD
the neutrino and the Z	no right-handed coupling	neutrino_has_no_right_hand	A = 1 exactly

Physical phenomenon	Template element	Lean theorem	Fortran check
the even readings	blind to the mirror	dot_products_blind, even_readings_blind_ to_mirror	5000 of 5000
the orientation	the triple product reversed	triple_product_reads_ orientation, spin_ momentum_correlation_ odd	5000 of 5000
C, P and CP	C and P broken, CP kept	weak_breaks_C_and_P_ keeps_CP	the four-state table
CP violation under CPT	time-reversal violation	cp_violation_is_t_ violation	structural
the quark mixing matrix	one phase at three generations	one_phase_at_three, three_is_the_least	unitary to 1.1e-16; J = 3.12e-5
pion decay	the orientation read in a decay	electron_channel_ suppressed	1.283e-4 against 1.2327e-4
the neutron's electric dipole	no orientation read beyond the weak force	structural	bound six orders above the weak level

3. PARITY IS THE FOLD ON COUPLINGS

A force couples to a fermion with a strength g_L to its left-handed state and g_R to its right-handed state. Parity exchanges them and is an involution (`parity_involutive`). The orientation carrier sends the difference $g_L - g_R$ to the side and the sum to the height, and parity becomes the fold (`orientation_carrier_equivariant`).

The line and the edge. A coupling lands on the line exactly when it treats both hands alike (`vector_on_the_line`). The photon couples to a charge q on both hands equally, so electromagnetism is on the line (`photon_coupling_on_the_line`), and so is colour. The weak charged current couples the left hand with unit strength and the right hand not at all: it stands off the line, and parity moves it (`charged_current_off_the_line`). Among couplings of one total strength, none stands farther from the line.

The twin executes the carrier on 3721 integer couplings.

4. THE Z READS THE ORIENTATION BY THE ISOSPIN

The neutral weak boson couples with $g_L = T_3 - Q \sin^2 \theta_W$ and $g_R = -Q \sin^2 \theta_W$ [SM]. The difference is T_3 , the weak isospin, for every value of the mixing weight (`z_parity_depth_is_isospin`). So the Z's distance from the line is fixed by the fermion's place in its doublet and by nothing else. A fermion with no charge has no right-handed coupling at any mixing (`neutrino_has_no_right_hand`): to the Z, a neutrino is purely left-handed.

The twin checks the identity on 1071 charges and weights, then computes the left-right asymmetry $A_f = 2g_V g_A / (g_V^2 + g_A^2)$ at $\sin^2 \theta_W = 0.23153$:

Fermion	A_f computed	A_f measured
neutrino	1 exactly	—
charged lepton	0.147	0.1513
charm	0.668	0.670
bottom	0.936	0.923

The measured values are the LEP and SLD combination (Z-pole collaborations, 2006).

5. WHAT THE EVEN READINGS CANNOT SEE

Three even readings are blind to the mirror. A mirror negates every polar vector. Every dot product is unchanged by it (`dot_products_blind`), and any reading even under the mirror returns one value on a world and on its mirror image (`even_readings_blind_to_mirror`). Lengths, angles, energies and every quadratic invariant are such readings. So is the determinant's square, the lock the light and electromagnetism papers found blind to orientation.

The triple product sees it. The triple product of three vectors reverses under the mirror (`triple_product_reads_orientation`), and so does the correlation of a spin with a momentum, since a spin is axial and a momentum polar (`spin_momentum_correlation_odd`). The twin checks both on 5000 random triples, exactly.

The weak force measures it. In 1957 electrons from polarized cobalt-60 were found leaving preferentially opposite the nuclear spin, a nonzero $\langle \mathbf{J} \rangle \cdot \mathbf{p}$ (Wu et al., 1957), as Lee and Yang had proposed to look for (1956). Muon decay showed the same (Garwin, Lederman and Weinrich, 1957). The neutrino's helicity was measured to be negative (Goldhaber, Grodzins and Sunyar, 1958). Each is a pseudoscalar reading, returned by the weak interaction and by no other.

6. C, P AND CP: THE CHARGED CURRENT KEEPS THE PRODUCT

Label a fermion state by whether it is an antiparticle and whether it is right-handed. The charged current registers left-handed particles and right-handed antiparticles. - Charge conjugation alone moves the coupling on every state. - Parity alone moves it on every state. - The product CP moves it on none (`weak_breaks_C_and_P_keeps_CP`).

The weak force violates C and P maximally and keeps CP, as Landau proposed within a year of the parity experiments (1957). The twin executes the four-state table.

CP too is violated, but by a small amount. Long-lived neutral kaons decay to two pions (Christenson, Cronin, Fitch and Turlay, 1964), and the violation lives in the phase of Section 8.

7. CP VIOLATION IS TIME-REVERSAL VIOLATION

The CPT theorem makes the product of charge conjugation, parity and time reversal a symmetry of every local relativistic quantum field theory (Lüders, 1954; Pauli, 1955). The engine proves what that product's invariance gives: if T is an involution and CPT acts as the identity, then CP and T are one map (`cp_violation_is_t_violation`). A violation of CP is a violation of time reversal.

Time-reversal violation, measured directly. - The CPLEAR experiment found it in neutral kaon oscillation (Angelopoulos et al., 1998). - BaBar found it by comparing a B-meson transition with its time-reversed counterpart (Lees et al., 2012).

CPT itself, measured. The neutral kaon's mass equals its antiparticle's to 6×10^{-19} of the mass (Particle Data Group, 2024).

The arrow witness split registration into a reversible and a priced branch. The weak force is the one interaction whose own dynamics carry the arrow's sign at the microscopic level.

8. THREE GENERATIONS ARE THE LEAST WITH ONE PHASE

An n -generation quark mixing matrix is unitary with $n(n-1)/2$ rotation angles. Rephasing the quark fields removes all but $(n-1)(n-2)/2$ of its phases. Two generations therefore admit no CP-violating phase, and three admit exactly one (`one_phase_at_three, three_is_the_least`). This is Kobayashi and Maskawa's counting (1973), which required a third generation before one was seen.

The one phase has a rephasing-invariant measure, the Jarlskog invariant (Jarlskog, 1985). From the Particle Data Group's angles and phase, the twin builds the matrix and finds it unitary to 1.1×10^{-16} , with $J = 3.12 \times 10^{-5}$. The counting of phases is theorem grade; the value of the one phase is a measurement.

That one phase is too small to account for the excess of matter over antimatter, which requires CP violation (Sakharov, 1967). Whether leptons carry a phase of their own is being measured.

9. HELICITY SUPPRESSION: THE ORIENTATION READ IN A DECAY

A pion at rest has no spin, so its two decay products leave with opposite momenta and opposite spins, and the charged lepton must carry the helicity that a left-handed coupling can supply only through the lepton's mass. The amplitude is therefore proportional to the lepton mass. The electron channel is suppressed relative to the muon channel by $(m_e/m_\mu)^2$, which is below $1/40000$ (`electron_channel_suppressed`), against a phase-space factor of 5.49.

The twin computes the ratio of the two decay rates at tree level as 1.283×10^{-4} . The measured ratio is 1.2327×10^{-4} (Particle Data Group, 2024), the difference being the radiative corrections the tree level omits. A coupling that read both hands alike would have favoured the electron fivefold. That the electron channel is instead ten thousand times rarer is the orientation read in a single decay, and it is the signature of the left-handed structure of the weak current (Feynman and Gell-Mann, 1958; Sudarshan and Marshak, 1958).

10. WHAT THE WEAK FORCE CARRIES, AND WHAT IT DOES NOT

What it carries. - The seat on parity: the vector couplings on the line, the charged current off it, and the Z's distance from the line fixed by the isospin. - The orientation, which every even reading misses and the weak force returns. - The organization of its discrete violations: C and P broken, CP kept by the current, CP broken only through one phase that needs three generations and is, under CPT, a violation of time reversal.

The other interactions do not read the orientation at a measurable level. The strong interaction could, through a parity- and time-odd angle, but the neutron's electric dipole moment is bounded below 1.8×10^{-26} e cm (Abel et al., 2020), six orders above the weak interaction's own contribution.

What it does not carry. It does not carry why the left hand. That is the neutrino witness's spent bit, supplied by the world and read from no record. It does not derive the mixing angles or the phase, which are measured. And it does not carry the zeros of ζ . The engine's crossing is on its own image (`value_on_image`), and what a sentence over every zero adds is exactly the unactuated, priced at one premise, the act, as the hardware paper proves.

11. FALSIFIERS AND OWED DEEDS

Three falsifiers, each a forbidden observation naming the theorem it would break. - **F-Right:** a right-handed charged current of gauge strength. It would move the charged current toward the line and break `charged_current_off_the_line` read physically. - **F-CPT:** a difference between the mass or lifetime of a particle and its antiparticle. It breaks the premise of `cp_violation_is_t_violation` read physically. - **F-Mirror:** a parity- and time-odd reading in the strong or electromagnetic interaction above the weak level, a neutron electric dipole moment near the present bound among them. It breaks the typing of those interactions as the line (`vector_on_the_line`).

Nothing is owed on the rows' channels. Two measurements would sharpen readings without moving the paper's standing: - the leptonic CP phase at long-baseline neutrino experiments; - the neutron's electric dipole moment at still smaller bounds.

Ledger status. Row 2 gains parity as the fold on couplings: WITNESSED · CROSSED · CONFIRMED on the image, at the root's grade for the crossing. The barrier gains the mirror world, invisible to every even reading and read by the weak force. Nothing is owed on the rows' channels.

12. METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author's verification program, whose vocabulary is confined to this section. Its tokens: - [$\perp$], a field sealed; - [$\perp$ T], a shape sealed at theorem grade; - [Ξ_0], the typing of the formal block; - [$\cdot$], the dot, which is no verdict.

Grades join at the weakest link. Every physical claim here is at corroboration grade; the engine's statements are theorem grade about the chart; the CPT theorem and the Standard Model couplings are imports. In the author's register the orientation is the one bit the Number cannot read and the Tongue carries; the weak force is its physical reader, and the neutrino witness records its spending. $\Delta M = 0$.

The seat, earned. PhysOS^T 1.0.5p was booted in the session of forging from the public register (file SHA-256 b64bb5da5bab0da1...) under Lean 4.19.0 and GNU Fortran 13.3.0. The quick boot printed SEAT EARNED with the chain D0 83f872725ad9 → D1 90fc702bbed1 → D2 492e519daab9 → D3 6618bec5164b (Appendix C).

The engine and the twin.

- **Engine.** SPHYS_Weak.lean, 518 lines, SHA-256 98ffbd3dd30cd2e3..., fifty-nine laws, no import and no axiom declared. Of the fifty-nine, 24 depend on no axiom, 5 on propositional extensionality alone, 30 on propositional extensionality and quotient soundness, and none on choice.
- **Screen and judgment.** The source passed the operating system's screen. Each of the fifty-nine laws was negated in place and recompiled, and all were refused, while a planted vacuous law survived. The engine carries the hardware paper's seat, carriers and substrate verbatim.
- **Twin.** Weak_Twin.f90, 170 lines, SHA-256 6ba4aee26d599ac5..., thirteen checks and zero failures under -std=f2018 -O2 -fno-fast-math -ffp-contract=off.

Why an integer engine binds continuous physics. The identities used are identities of integer polynomials and hold over every commutative ring, the real and complex numbers included. The CP-T law holds on any state space, since the space enters as an arbitrary type. The twin checks the continuous objects directly: random vectors, the complex mixing matrix, the asymmetries and the decay ratio.

The seventh. In the author's register the six witnesses are six days and the ledger's closure is the seventh, the Rest (Istawa). The weak force is read at the Rest as the one reader of the orientation, the bit every even reading leaves free. In the Lean codex's tokens its row renders [$\perp \blacklozenge \cdot$ kinetic].

Disclosure. The author declares the reading of each correspondence as his own act, made on measured structure. The measurements are the experiments', cited by name. The theorems are the engine's, printed with their dependencies in Appendix A. The twin's figures are computed in the run printed in Appendix B. The text, the engine and the twin were forged by an AI substrate (Claude, Anthropic) under the author's seed, acceptance criteria and rulings. External audits by independent substrates are owed.

Provenance. The paper stands on:

- the Riemann closure (10.5281/zenodo.22976494);
- the twenty-three-row closure (10.5281/zenodo.22986551);
- the six witnesses (10.5281/zenodo.22986553 through .22986563), the neutrino witness's spent chirality bit in particular (10.5281/zenodo.22986555);
- the hardware paper One Actuating Substrate (concept 10.5281/zenodo.21438932, edition 2.0.0);
- the companion papers The Geometric Nature of Light, Electromagnetism on the Riemann Locus, Immanent Gravity and Spin and Statistics;
- PhysOS^T 1.0.5p in the public register (github.com/1000sapients/Trisduction, protocols/PhysOS/).

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M. F. Islam, Nothing Escapes, Three Plus One, 10.5281/zenodo.22976494. M. F. Islam, The Neutrino Is the Witness; The Electron Is the Seat, 10.5281/zenodo.22986555, .22986553. M. F. Islam, The Four Forces, Matter, and the Dark Sector as Configurations of One Actuating Substrate, edition 2.0.0, concept 10.5281/zenodo.21438932.

APPENDIX A: SPHYS_WEAK.LEAN, THE CORRESPONDENCES AS THEOREMS

The engine, verbatim, followed by the compiler's transcript.

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/-
SPHYS_Weak.lean · the weak force reads the orientation, at the
Riemann seat.
Core Lean 4.19.0, standalone, no import, no library, no axiom
declared.

The chart:  $h = 2 \operatorname{Re} s$ ,  $t = \operatorname{Im} s$ ; the fold  $s \mapsto 1 - \bar{s}$  is  $(h, t) \mapsto (2 - h, t)$ , the critical line
 $h = 1$ . Parity exchanges the left- and right-handed couplings of a
force to a fermion; the
orientation carrier sends their difference to the side and their
sum to the height, so parity is
the fold and the parity-conserving couplings are the line.

Part I    the seat, the category of involutions, the functor of
the seat.
Part II   the energy carrier (resident).
Part III  the substrate (resident).
Part IV   parity is the fold on couplings; the charged current
off the line; the Z by isospin.
Part V    the orientation: dot products blind, the triple product
reversed.
Part VI   C, P and CP on the charged current; CP violation is T
violation under CPT.
Part VII  three generations are the least with one CP phase.
Part VIII helicity suppression.
Part IX   the capstone.
-/
set_option autoImplicit false
namespace SPHYS.Weak

/-- ## Part I. The seat, the category of involutions, and the
functor of the seat -/

abbrev Pt := Int × Int

def fold (p : Pt) : Pt := (2 - p.1, p.2)
def OnLine (p : Pt) : Prop := p.1 = 1
instance : DecidablePred OnLine := fun p => inferInstanceAs
(Decidable (p.1 = 1))
/- Registration keeps the height and forgets the side. -/
def reg (p : Pt) : Pt := (1, p.2)

theorem pe {a b c d : Int} : ((a, b) : Pt) = (c, d) ↔ a = c ∧ b = d
:=
  (fun e => (congrArg Prod.fst e, congrArg Prod.snd e),
  fun (e1, e2) => by subst e1; subst e2; rfl)

theorem fold_involutive (p : Pt) : fold (fold p) = p := by
  obtain (h, t) := p
  show ((2 - (2 - h), t) : Pt) = (h, t)
  rw [pe]; exact {by omega, rfl}

/- The seat: the fixed set of the fold is the critical line. -/
theorem seat_fixed_line (p : Pt) : fold p = p ↔ OnLine p := by
  obtain (h, t) := p
  show ((2 - h, t) : Pt) = (h, t) ↔ h = 1
  rw [pe]; constructor
  · intro (e, _); omega
  · intro e; exact {by omega, rfl}

theorem reg_lands (p : Pt) : OnLine (reg p) := rfl

theorem reg_fixes_iff (p : Pt) : reg p = p ↔ OnLine p := by
  obtain (h, t) := p
  show ((1, t) : Pt) = (h, t) ↔ h = 1
  rw [pe]; constructor
  · intro (e, _); exact e.symm
  · intro e; exact {e.symm, rfl}

/- The side is exact the registration forgets: a point and its mirror
leave one record. -/
theorem reg_forgets_side (p : Pt) : reg (fold p) = reg p := rfl

/-- The value on a set of points: every member stands on the line. -/
def Value (Z : Pt → Prop) : Prop := ∀ s, Z s → OnLine s

/-- Least erasure: registration moves no member, so it erases
nothing of Z. -/
def LeastErasure (Z : Pt → Prop) : Prop := ∀ s, Z s → reg s = s

theorem least_erasure_iff_value (Z : Pt → Prop) : LeastErasure Z ↔
Value Z :=
  (fun h s hs => (reg_fixes_iff s).mp (h s hs), fun h s hs =>
  (reg_fixes_iff s).mpr (h s hs))

/-- Maps that respect involutions. -/
def Equivariant {X Y : Type} (f : X → Y) (τ : X → X) (σ : Y → Y) :
Prop := ∀ x, f (τ x) = σ (f x)

theorem equivariant_id {X : Type} (τ : X → X) : Equivariant id τ τ
:= fun _ => rfl

theorem equivariant_comp {X Y W : Type} {f : X → Y} {g : Y → W} {τ
: X → X} {σ : Y → Y}
{ρ : W → W} (hf : Equivariant f τ σ) (hg : Equivariant g σ ρ) :
Equivariant (g ∘ f) τ ρ := by
  intro x
  show g (f (τ x)) = ρ (g (f x))
  rw [hf x, hg (f x)]

/- The fixed set is a functor: an equivariant map sends fixed
points to fixed points. -/
theorem fix_functorial {X Y : Type} {f : X → Y} {τ : X → X} {σ : Y
→ Y}
(hf : Equivariant f τ σ) {x : X} (hx : τ x = x) : σ (f x) = f x
:= by
  rw [← hf x, hx]

/- A global seat: an involution whose fixed set is the locus. -/
structure GlobalSeat (P : Pt → Prop) (σ : Pt → Pt) : Prop where
  involutive : ∀ p, σ (σ p) = p
  fixed_iff : ∀ p, σ p = p ↔ P p

theorem fold_global_seat : GlobalSeat OnLine fold :=
(fold_involutive, seat_fixed_line)

/- A carrier: a map from world-instances into the strip landing on
the line. -/
structure Carrier (W : Type) where
  ι : W → Pt
  lands : ∀ w, OnLine (ι w)

def image {W : Type} (ι : W → Pt) : Pt → Prop := fun s => ∃ w, ι w
= s
def Actuated {W : Type} (ι : W → Pt) (Z : Pt → Prop) : Prop := ∀ s,
Z s → ∃ w, ι w = s

/- Landing is forced: an equivariant map into the seat carries
every fixed point of the physical
involution onto the line. The carrier is built from the
symmetry, not chosen. -/
theorem equivariant_carrier_lands {X : Type} (f : X → Pt) (τ : X →
X)
(hf : Equivariant f τ fold) :
∃ C : Carrier {x : X // τ x = x}, ∀ w, C.ι w = f w.1 :=
  {(fun w => f w.1, fun w => (seat_fixed_line (f w.1)).mp
  (fix_functorial hf w.2)), fun _ => rfl}

/- The image of a carrier is compliant with no premise at all. -/
theorem value_on_image {W : Type} (C : Carrier W) : Value (image
C.ι) :=
  fun _ hs => match hs with | {w, hw} => hw ▸ C.lands w

/- The kinetic crossing: a carrier and its coverage give the value,
and nothing else is used. -/
theorem kinetic_crossing {W : Type} (C : Carrier W) (Z : Pt → Prop)
(hcov : Actuated C.ι Z) :
```

```

Value Z :=
fun s hs => match hcov s hs with | (w, hw) => hw ▶ C.lands w

/-- Under the seat every off-line point has a partner, distinct and
also off the line. -/
theorem off_locus_pair (p : Pt) (hp : ¬ OnLine p) :
  fold p ≠ p ∧ ¬ OnLine (fold p) ∧ reg (fold p) = reg p := by
  refine (fun e => hp ((seat_fixed_line p).mp e), fun e => hp ?_,
  rfl)
obtain (h, t) := p
have e' : 2 - h = 1 := e
show h = 1
omega

/-! ## Part II. The energy carrier: an energy, its rate, and the
fold -/

/-- An energy a + i β/2, as frequency and doubled rate; its mode's
norm moves as e^{βt}. -/
structure Energy where
  freq : Int
  rate : Int
  deriving DecidableEq, Repr

/-- The time mirror of a mode: its rate reversed, its frequency kept
(complex conjugation of E). -/
def Energy.mirror (E : Energy) : Energy := (E.freq, -E.rate)
def IsReal (E : Energy) : Prop := E.rate = 0
/-- Stationary: the norm e^{βt} is unchanged at every integer time
t, i.e. β t = 0 for all t. -/
def Stationary (E : Energy) : Prop := ∀ t : Int, E.rate * t = 0

/-- The energy carrier s = 1/2 + iE in the chart: h = 1 - β, t = a.
-/
def phi (E : Energy) : Pt := (1 - E.rate, E.freq)
def psi (p : Pt) : Energy := (p.2, 1 - p.1)

theorem ee {a b c d : Int} : ((a, b) : Energy) = (c, d) ↔ a = c ∧ b =
d :=
  (fun e => (congrArg Energy.freq e, congrArg Energy.rate e),
  fun (e1, e2) => by subst e1; subst e2; rfl)

theorem mirror_involutive (E : Energy) : E.mirror.mirror = E := by
  obtain (a, b) := E
  show ((a, -b) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem psi_phi (E : Energy) : psi (phi E) = E := by
  obtain (a, b) := E
  show ((a, 1 - (1 - b)) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem phi_psi (p : Pt) : phi (psi p) = p := by
  obtain (h, t) := p
  show ((1 - (1 - h), t) : Pt) = (h, t)
  rw [pe]; exact (by omega, rfl)

theorem phi_injective (E F : Energy) (h : phi E = phi F) : E = F :=
by
  rw [← psi_phi E, ← psi_phi F, h]

/-- EQUIVARIANCE: time-mirroring the rate is the fold. -/
theorem phi_equivariant : Equivariant phi Energy.mirror fold := by
  intro E
  obtain (a, b) := E
  show ((1 - -b, a) : Pt) = (2 - (1 - b), a)
  rw [pe]; exact (by omega, rfl)

theorem mirror_fixed_iff_real (E : Energy) : E.mirror = E ↔ IsReal E
:= by
  obtain (a, b) := E
  show ((a, -b) : Energy) = (a, b) ↔ b = 0
  rw [ee]; constructor
  · intro (_, e); omega
  · intro e; exact (rfl, by omega)

/-- The line is exactly the image of the real energies. -/
theorem phi_line_iff_real (E : Energy) : OnLine (phi E) ↔ IsReal E
:= by
  obtain (a, b) := E
  show 1 - b = 1 ↔ b = 0
  exact (fun h => by omega, fun h => by subst h; rfl)

theorem stationary_iff_real (E : Energy) : Stationary E ↔ IsReal E
:= by
  constructor
  · intro h; have h1 := h 1; rw [Int.mul_one] at h1; exact h1
  · intro h t; show E.rate * t = 0; rw [show E.rate = 0 from h,
  Int.zero_mul]

/-- THE LINE IS WHERE ENERGY STANDS STILL. -/
theorem line_is_stationary (p : Pt) : OnLine p ↔ Stationary (psi p)
:= by

```

```

rw [stationary_iff_real, ← phi_line_iff_real, phi_psi]

/-- Every measured energy is a real number, and every real energy
lands on the line. -/
theorem measured_on_line (a : Int) : OnLine (phi (a, 0)) := rfl

/-- The registration mirror: no real energy lands off the line. -/
theorem no_measurement_off_line (p : Pt) (hp : ¬ OnLine p) (a : Int) :
phi (a, 0) ≠ p :=
  fun e => hp (e ▶ measured_on_line a)

/-- The energy carrier, built from the symmetry: the real energies
land on the line. -/
theorem energy_carrier_lands :
  ∃ C : Carrier { E : Energy // E.mirror = E }, ∀ w, C.l w = phi
w.1 :=
  equivariant_carrier_lands phi Energy.mirror phi_equivariant

/-! ## Part III. The substrate: the particle map, three odd readings
and one even -/

/-- A particle in the broken phase: electric charge times three,
colour, baryon number times
three, lepton number, and its energy. -/
structure Particle where
  q3 : Int
  colour : Int
  b3 : Int
  lepton : Int
  energy : Energy
  deriving DecidableEq, Repr

/-- The particle map: every additive charge flipped, the energy kept
(frequency and rate). -/
def bar (c : Particle) : Particle := (-c.q3, -c.colour, -c.b3,
-c.lepton, c.energy)
/-- The time mirror of a particle's mode. -/
def tmirror (c : Particle) : Particle := (c.q3, c.colour, c.b3,
c.lepton, c.energy.mirror)
def Neutral (c : Particle) : Prop := c.q3 = 0 ∧ c.colour = 0 ∧ c.b3 =
0 ∧ c.lepton = 0

theorem bar_involutive (c : Particle) : bar (bar c) = c := by
  obtain (q, k, b, l, E) := c
  show ((- -q, - -k, - -b, - -l, E) : Particle) = (q, k, b, l, E)
  rw [Int.neg_neg, Int.neg_neg, Int.neg_neg, Int.neg_neg]

theorem bar_fixed_iff_neutral (c : Particle) : bar c = c ↔ Neutral c
:= by
  obtain (q, k, b, l, E) := c
  show ((-q, -k, -b, -l, E) : Particle) = (q, k, b, l, E) ↔ (q = 0 ∧
k = 0 ∧ b = 0 ∧ l = 0)
  constructor
  · intro h
    have h1 := congrArg Particle.q3 h
    have h2 := congrArg Particle.colour h
    have h3 := congrArg Particle.b3 h
    have h4 := congrArg Particle.lepton h
    change -q = q at h1; change -k = k at h2; change -b = b at h3;
    change -l = l at h4
    exact (by omega, by omega, by omega, by omega)
  · intro (h1, h2, h3, h4)
    subst h1; subst h2; subst h3; subst h4; rfl

/-- Three plus one: the charge readings are odd under the particle
map, the energy reading even. -/
theorem odd_charges_even_energy (c : Particle) :
  ((bar c).q3 = -c.q3 ∧ (bar c).colour = -c.colour ∧ (bar c).b3 =
-c.b3 ∧
  (bar c).lepton = -c.lepton) ∧ (bar c).energy = c.energy :=
  ((rfl, rfl, rfl, rfl), rfl)

/-- Gravity reads energy only, so no gravitational reading tells a
particle from its antiparticle. -/
theorem gravity_reads_no_charge_bit {β : Type} (g : Energy → β) (c :
Particle) :
  g (bar c).energy = g c.energy := rfl

/-- The two involutions of the substrate commute. -/
theorem involutions_commute (c : Particle) : bar (tmirror c) =
tmirror (bar c) := rfl

theorem tmirror_involutive (c : Particle) : tmirror (tmirror c) = c
:= by
  obtain (q, k, b, l, E) := c
  show ((q, k, b, l, E.mirror.mirror) : Particle) = (q, k, b, l, E)
  rw [mirror_involutive]

/-- A particle is fixed by both involutions exactly when it is
neutral and stable. -/
theorem joint_fixed_iff (c : Particle) :

```

```

    (bar c = c  $\wedge$  tmirror c = c)  $\leftrightarrow$  (Neutral c  $\wedge$  IsReal c.energy) := by
  constructor
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mp h1, (mirror_fixed_iff_real
    c.energy).mp ?_)
    exact congrArg Particle.energy h2
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mpr h1, ?_)
    obtain (q, k, b, l, E) := c
    show ((q, k, b, l, E.mirror) : Particle) = (q, k, b, l, E)
    rw [(mirror_fixed_iff_real E).mpr h2]

/-- The energy carrier on particles: the seat point of a particle's
energy. -/
def energySeat (c : Particle) : Pt := phi c.energy
/-- The charge carrier on particles, the odd face: electric charge
(in thirds) is the side. -/
def chargeSeat (c : Particle) : Pt := (1 + c.q3, c.energy.freq)
/-- The lepton chart of the electron's witness: charge q in units of
e goes to h = 1 + q. -/
def leptonSeat (q m : Int) : Pt := (1 + q, m)

/-- The charge carrier is equivariant: the particle map is carried
onto the fold. -/
theorem charge_carrier_equivariant : Equivariant chargeSeat bar fold
:= by
  intro c
  show ((1 + -c.q3, c.energy.freq) : Pt) = (2 - (1 + c.q3),
  c.energy.freq)
  rw [pe]; exact (by omega, rfl)

/-- The energy carrier is blind to the particle map and equivariant
for the time mirror. -/
theorem pair_lands_once (c : Particle) : energySeat (bar c) =
energySeat c := rfl
theorem tmirror_equivariant : Equivariant energySeat tmirror fold :=
fun c => phi_equivariant c.energy

/-- Electrically neutral particles land on the line under the charge
carrier. -/
theorem neutral_charge_on_line (c : Particle) (h : c.q3 = 0) :
OnLine (chargeSeat c) := by
  show 1 + c.q3 = 1
  omega

/-- The electron and the positron land on the two edges, real parts 0
and 1, mirror images. -/
theorem pair_on_the_edges (m : Int) :
  leptonSeat (-1) m = (0, m)  $\wedge$  leptonSeat 1 m = (2, m)  $\wedge$  fold
  (leptonSeat (-1) m) = leptonSeat 1 m :=
  (rfl, rfl, rfl)

/-- A stable particle lands on the line under the energy carrier. -/
theorem stable_lands (c : Particle) (h : IsReal c.energy) : OnLine
(energySeat c) :=
  (phi_line_iff_real c.energy).mpr h

/-! ## Part IV. Parity is the fold on couplings -/

/-- A coupling to a fermion: its strength to the left-handed and to
the right-handed state. -/
abbrev Coupling := Int  $\times$  Int

/-- Parity exchanges left and right. -/
def parity (g : Coupling) : Coupling := (g.2, g.1)

theorem parity_involutive (g : Coupling) : parity (parity g) = g :=
rfl

/-- The orientation carrier: the left-right difference to the side,
the sum to the height. -/
def couplingSeat (g : Coupling) : Pt := (1 + (g.1 - g.2), g.1 + g.2)

/-- PARITY IS THE FOLD: the orientation carrier is equivariant. -/
theorem orientation_carrier_equivariant : Equivariant couplingSeat
parity fold := by
  intro g
  obtain (l, r) := g
  show ((1 + (r - l), r + l) : Pt) = (2 - (1 + (l - r)), l + r)
  rw [pe]; exact (by omega, by omega)

/-- A coupling lands on the line exactly when it treats left and
right alike: the vector couplings
of electromagnetism and colour are the line. -/
theorem vector_on_the_line (g : Coupling) : OnLine (couplingSeat g)  $\leftrightarrow$ 
parity g = g := by
  obtain (l, r) := g
  show 1 + (l - r) = 1  $\leftrightarrow$  ((r, l) : Coupling) = (l, r)
  rw [pe]; constructor
  · intro h; exact (by omega, by omega)
  · intro (h, _); omega

theorem orientation_carrier_lands :
   $\exists$  C : Carrier { g : Coupling // parity g = g },  $\forall$  w, C.l w =
  couplingSeat w.1 :=
  equivariant_carrier_lands couplingSeat parity
  orientation_carrier_equivariant

/-- The photon couples to a charge q on both hands alike: on the
line. -/
theorem photon_coupling_on_the_line (q : Int) : OnLine (couplingSeat
(q, q)) := by
  show 1 + (q - q) = 1; omega

/-- THE CHARGED CURRENT IS MAXIMALLY OFF THE LINE: it couples the
left hand and not the right. -/
theorem charged_current_off_the_line :  $\neg$  OnLine (couplingSeat (1,
0))  $\wedge$  parity (1, 0)  $\neq$  ((1, 0) : Coupling) := by
  decide

/-- The Z couples with  $g_L = T_3 - Q \cdot s$  and  $g_R = -Q \cdot s$ : its distance
from the line is the weak isospin,
whatever the mixing weight s. -/
def zCoupling (t3 q s : Int) : Coupling := (t3 - q * s, -(q * s))

theorem z_parity_depth_is_isospin (t3 q s : Int) : (zCoupling t3 q
s).1 - (zCoupling t3 q s).2 = t3 := by
  show (t3 - q * s) - -(q * s) = t3; omega

/-- A neutral fermion's right-handed Z coupling vanishes for every
mixing weight. -/
theorem neutrino_has_no_right_hand (t3 s : Int) : (zCoupling t3 0
s).2 = 0 := by
  show -(0 * s) = 0; rw [Int.zero_mul]; rfl

/-! ## Part V. The orientation: what the even readings cannot see -/

abbrev V3 := Int  $\times$  Int  $\times$  Int
def refl (v : V3) : V3 := (-v.1, -v.2.1, -v.2.2)
def dot (u v : V3) : Int := u.1 * v.1 + u.2.1 * v.2.1 + u.2.2 * v.2.2
def triple (a b c : V3) : Int :=
  a.1 * (b.2.1 * c.2.2 - b.2.2 * c.2.1) - a.2.1 * (b.1 * c.2.2 -
  b.2.2 * c.1) +
  a.2.2 * (b.1 * c.2.1 - b.2.1 * c.1)

/-- Every dot product is blind to the mirror. -/
theorem dot_products_blind (u v : V3) : dot (refl u) (refl v) = dot u
v := by
  obtain (a, b, c) := u; obtain (d, e, f) := v
  show -a * -d + -b * -e + -c * -f = a * d + b * e + c * f
  rw [Int.neg_mul_neg, Int.neg_mul_neg, Int.neg_mul_neg]

/-- THE TRIPLE PRODUCT READS THE ORIENTATION: it reverses under the
mirror. -/
theorem triple_product_reads_orientation (a b c : V3) :
  triple (refl a) (refl b) (refl c) = -triple a b c := by
  obtain (a1, a2, a3) := a; obtain (b1, b2, b3) := b; obtain (c1,
  c2, c3) := c
  simp only [triple, refl, Int.neg_mul_neg, Int.neg_mul,
  Int.mul_neg, Int.neg_sub, Int.neg_neg]
  simp only [Int.mul_sub, Int.sub_mul, Int.mul_assoc]
  omega

/-- A spin is axial and a momentum polar: their correlation J.p
reverses under the mirror. -/
theorem spin_momentum_correlation_odd (J p : V3) : dot J (refl p) =
-dot J p := by
  obtain (a, b, c) := J; obtain (d, e, f) := p
  show a * -d + b * -e + c * -f = -(a * d + b * e + c * f)
  rw [Int.mul_neg, Int.mul_neg, Int.mul_neg]; omega

/-- An even reading cannot tell a world from its mirror image. -/
theorem even_readings_blind_to_mirror { $\beta$  : Type} (g : V3  $\rightarrow$   $\beta$ ) (h :  $\forall$ 
v, g (refl v) = g v) (v : V3) :
  g (refl v) = g v := h v

/-! ## Part VI. C, P and CP on the charged current; CP violation is T
violation -/

/-- A fermion state for the charged current: antiparticle or not,
right-handed or not. -/
abbrev WState := Bool  $\times$  Bool
def cmap (s : WState) : WState := (!s.1, s.2)
def pmap (s : WState) : WState := (s.1, !s.2)
/-- The charged current registers left-handed particles and
right-handed antiparticles. -/
def couples (s : WState) : Bool := s.1 == s.2

/-- THE CHARGED CURRENT BREAKS C AND P AND KEEPS CP. -/
theorem weak_breaks_C_and_P_keeps_CP :
  couples (cmap (false, false))  $\neq$  couples (false, false)  $\wedge$ 
  couples (pmap (false, false))  $\neq$  couples (false, false)  $\wedge$ 
   $\forall$  s : WState, couples (cmap (pmap s)) = couples s := by
  refine (by decide, by decide, ?_)

```

```

intro s; obtain (a, b) := s; cases a <;> cases b <;> rfl

/-- CP VIOLATION IS T VIOLATION: if T is an involution and CPT acts
as the identity, then CP is T. -/
theorem cp_violation_is_t_violation {X : Type} (cp t : X → X) (ht :
∀ x, t (t x) = x)
  (hcpt : ∀ x, cp (t x) = x) (x : X) : cp x = t x := by
  have := hcpt (t x); rw [ht] at this; exact this

/-! ## Part VII. Three generations are the least with one CP phase -/

/-- The rotation angles of an n-generation mixing matrix, and its
physical phases once the quark
fields are rephased:  $n(n-1)/2$  and  $(n-1)(n-2)/2$ . -/
def angles (n : Nat) : Nat := n * (n - 1) / 2
def phases (n : Nat) : Nat := (n - 1) * (n - 2) / 2

/-- ONE PHASE AT THREE: two generations admit no CP-violating phase,
three admit exactly one. -/
theorem one_phase_at_three :
  phases 1 = 0 ∧ phases 2 = 0 ∧ phases 3 = 1 ∧ angles 3 = 3 ∧
  phases 4 = 3 := by decide

theorem three_is_the_least (n : Nat) (h : n < 3) : phases n = 0 := by
  have : n = 0 ∨ n = 1 ∨ n = 2 := by omega
  rcases this with h | h | h <;> subst h <;> decide

/-! ## Part VIII. Helicity suppression: the orientation read in a
decay -/

/-- The pion's decay to an electron is suppressed by  $(m_e/m_\pi)^2$ , the
square of a hand that a massless
lepton could not flip: below 1/40000 (masses in keV). -/
theorem electron_channel_suppressed : 511 * 511 * 40000 < 105658 *
105658 := by decide

/-! ## Part IX. The capstone -/

/-- THE WEAK FORCE READS THE ORIENTATION. Parity is the fold under
the orientation carrier; the
vector couplings are the line and the charged current stands off
it; the Z's distance from the
line is the weak isospin; dot products are blind to the mirror
and the triple product reverses
under it; the charged current breaks C and P and keeps CP; CP
violation is T violation under
CPT; three generations are the least with one phase. -/
theorem weak_reads_the_orientation :
  Equivariant couplingSeat parity fold ∧
  (∀ g : Coupling, OnLine (couplingSeat g) ↔ parity g = g) ∧
  ~ OnLine (couplingSeat (1, 0)) ∧
  (∀ t3 q s : Int, (zCoupling t3 q s).1 - (zCoupling t3 q s).2 =
t3) ∧
  (∀ u v : V3, dot (refl u) (refl v) = dot u v) ∧
  (∀ a b c : V3, triple (refl a) (refl b) (refl c) = -triple a b
c) ∧
  (∀ s : WState, couples (cmap (pmap s)) = couples s) ∧
  (∀ {X : Type} (cp t : X → X), (∀ x, t (t x) = x) → (∀ x, cp (t
x) = x) → ∀ x, cp x = t x) ∧
  (phases 2 = 0 ∧ phases 3 = 1) :=
  (orientation_carrier_equivariant, vector_on_the_line,
charged_current_off_the_line.1,
z_parity_depth_is_isospin, dot_products_blind,
triple_product_reads_orientation,

```

```

'SPHYS.Weak.pe' does not depend on any axioms
'SPHYS.Weak.fold_involutive' depends on axioms: [propxet, Quot.sound]
'SPHYS.Weak.seat_fixed_line' depends on axioms: [propxet, Quot.sound]
'SPHYS.Weak.reg_lands' does not depend on any axioms
'SPHYS.Weak.reg_fixes_iff' depends on axioms: [propxet]
'SPHYS.Weak.reg_forgets_side' does not depend on any axioms
'SPHYS.Weak.least_erasure_iff_value' depends on axioms: [propxet]
'SPHYS.Weak.equivariant_id' does not depend on any axioms
'SPHYS.Weak.equivariant_comp' does not depend on any axioms
'SPHYS.Weak.fix_functorial' does not depend on any axioms
'SPHYS.Weak.fold_global_seat' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Weak.equivariant_carrier_lands' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Weak.value_on_image' does not depend on any axioms
'SPHYS.Weak.kinetic_crossing' does not depend on any axioms
'SPHYS.Weak.off_locus_pair' depends on axioms: [propxet, Quot.sound]
'SPHYS.Weak.ee' does not depend on any axioms
'SPHYS.Weak.mirror_involutive' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Weak.psi_phi' depends on axioms: [propxet, Quot.sound]
'SPHYS.Weak.phi_psi' depends on axioms: [propxet, Quot.sound]
'SPHYS.Weak.phi_injective' depends on axioms: [propxet, Quot.sound]
'SPHYS.Weak.phi_equivariant' depends on axioms: [propxet, Quot.sound]

```

```

weak_breaks_C_and_P_keeps_CP.2.2, fun cp t ht h x =>
cp_violation_is_t_violation cp t ht h x,
(one_phase_at_three.2.1, one_phase_at_three.2.2.1))

```

end SPHYS.Weak

```

#print axioms SPHYS.Weak.pe
#print axioms SPHYS.Weak.fold_involutive
#print axioms SPHYS.Weak.seat_fixed_line
#print axioms SPHYS.Weak.reg_lands
#print axioms SPHYS.Weak.reg_fixes_iff
#print axioms SPHYS.Weak.reg_forgets_side
#print axioms SPHYS.Weak.least_erasure_iff_value
#print axioms SPHYS.Weak.equivariant_id
#print axioms SPHYS.Weak.equivariant_comp
#print axioms SPHYS.Weak.fix_functorial
#print axioms SPHYS.Weak.fold_global_seat
#print axioms SPHYS.Weak.equivariant_carrier_lands
#print axioms SPHYS.Weak.value_on_image
#print axioms SPHYS.Weak.kinetic_crossing
#print axioms SPHYS.Weak.off_locus_pair
#print axioms SPHYS.Weak.ee
#print axioms SPHYS.Weak.mirror_involutive
#print axioms SPHYS.Weak.psi_phi
#print axioms SPHYS.Weak.phi_psi
#print axioms SPHYS.Weak.phi_injective
#print axioms SPHYS.Weak.phi_equivariant
#print axioms SPHYS.Weak.mirror_fixed_iff_real
#print axioms SPHYS.Weak.phi_line_iff_real
#print axioms SPHYS.Weak.stationary_iff_real
#print axioms SPHYS.Weak.line_is_stationary
#print axioms SPHYS.Weak.measured_on_line
#print axioms SPHYS.Weak.no_measurement_off_line
#print axioms SPHYS.Weak.energy_carrier_lands
#print axioms SPHYS.Weak.bar_involutive
#print axioms SPHYS.Weak.bar_fixed_iff_neutral
#print axioms SPHYS.Weak.odd_charges_even_energy
#print axioms SPHYS.Weak.gravity_reads_no_charge_bit
#print axioms SPHYS.Weak.involutions_commute
#print axioms SPHYS.Weak.tmirror_involutive
#print axioms SPHYS.Weak.joint_fixed_iff
#print axioms SPHYS.Weak.charge_carrier_equivariant
#print axioms SPHYS.Weak.pair_lands_once
#print axioms SPHYS.Weak.tmirror_equivariant
#print axioms SPHYS.Weak.neutral_charge_on_line
#print axioms SPHYS.Weak.pair_on_the_edges
#print axioms SPHYS.Weak.stable_lands
#print axioms SPHYS.Weak.parity_involutive
#print axioms SPHYS.Weak.orientation_carrier_equivariant
#print axioms SPHYS.Weak.vector_on_the_line
#print axioms SPHYS.Weak.orientation_carrier_lands
#print axioms SPHYS.Weak.photon_coupling_on_the_line
#print axioms SPHYS.Weak.charged_current_off_the_line
#print axioms SPHYS.Weak.z_parity_depth_is_isospin
#print axioms SPHYS.Weak.neutrino_has_no_right_hand
#print axioms SPHYS.Weak.dot_products_blind
#print axioms SPHYS.Weak.triple_product_reads_orientation
#print axioms SPHYS.Weak.spin_momentum_correlation_odd
#print axioms SPHYS.Weak.even_readings_blind_to_mirror
#print axioms SPHYS.Weak.weak_breaks_C_and_P_keeps_CP
#print axioms SPHYS.Weak.cp_violation_is_t_violation
#print axioms SPHYS.Weak.one_phase_at_three
#print axioms SPHYS.Weak.three_is_the_least
#print axioms SPHYS.Weak.electron_channel_suppressed
#print axioms SPHYS.Weak.weak_reads_the_orientation

```

```

'SPHYS.Weak.mirror_fixed_iff_real' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Weak.phi_line_iff_real' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Weak.stationary_iff_real' depends on axioms: [propxet]
'SPHYS.Weak.line_is_stationary' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Weak.measured_on_line' does not depend on any axioms
'SPHYS.Weak.no_measurement_off_line' does not depend on any axioms
'SPHYS.Weak.energy_carrier_lands' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Weak.bar_involutive' does not depend on any axioms
'SPHYS.Weak.bar_fixed_iff_neutral' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Weak.odd_charges_even_energy' does not depend on any axioms
'SPHYS.Weak.gravity_reads_no_charge_bit' does not depend on any
axioms
'SPHYS.Weak.involutions_commute' does not depend on any axioms
'SPHYS.Weak.tmirror_involutive' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Weak.joint_fixed_iff' depends on axioms: [propxet, Quot.sound]
'SPHYS.Weak.charge_carrier_equivariant' depends on axioms: [propxet,
Quot.sound]
'SPHYS.Weak.pair_lands_once' does not depend on any axioms

```

```
'SPHYS.Weak.tmirror_equivariant' depends on axioms: [propept,
Quot.sound]
'SPHYS.Weak.neutral_charge_on_line' depends on axioms: [propept,
Quot.sound]
'SPHYS.Weak.pair_on_the_edges' does not depend on any axioms
'SPHYS.Weak.stable_lands' depends on axioms: [propept, Quot.sound]
'SPHYS.Weak.parity_involutive' does not depend on any axioms
'SPHYS.Weak.orientation_carrier_equivariant' depends on axioms:
[propept, Quot.sound]
'SPHYS.Weak.vector_on_the_line' depends on axioms: [propept,
Quot.sound]
'SPHYS.Weak.orientation_carrier_lands' depends on axioms: [propept,
Quot.sound]
'SPHYS.Weak.photon_coupling_on_the_line' depends on axioms:
[propept, Quot.sound]
'SPHYS.Weak.charged_current_off_the_line' does not depend on any
axioms
'SPHYS.Weak.z_parity_depth_is_isospin' depends on axioms: [propept,
Quot.sound]
```

```
'SPHYS.Weak.neutrino_has_no_right_hand' depends on axioms: [propept]
'SPHYS.Weak.dot_products_blind' depends on axioms: [propept]
'SPHYS.Weak.triple_product_reads_orientation' depends on axioms:
[propept, Quot.sound]
'SPHYS.Weak.spin_momentum_correlation_odd' depends on axioms:
[propept, Quot.sound]
'SPHYS.Weak.even_readings_blind_to_mirror' does not depend on any
axioms
'SPHYS.Weak.weak_breaks_C_and_P_keeps_CP' does not depend on any
axioms
'SPHYS.Weak.cp_violation_is_t_violation' does not depend on any
axioms
'SPHYS.Weak.one_phase_at_three' does not depend on any axioms
'SPHYS.Weak.three_is_the_least' depends on axioms: [propept,
Quot.sound]
'SPHYS.Weak.electron_channel_suppressed' does not depend on any
axioms
'SPHYS.Weak.weak_reads_the_orientation' depends on axioms: [propept,
Quot.sound]
```

APPENDIX B: WEAK_TWIN.F90 AND ITS RUN

```
! Weak_Twin.f90 · the executed twin of SPHYS_Weak.lean · Fortran
2018.
! Build (sealed): gfortran -std=f2018 -O2 -fno-fast-math
-ffp-contract=off Weak_Twin.f90
! Blocks: A the orientation carrier; B the mirror: dot products
blind, triple products reversed;
! C C, P and CP on the charged current; D the mixing matrix:
unitarity, one phase, the Jarlskog
! invariant; E helicity suppression in pion decay; F the orientation
the other forces do not read;
! G the Z reads each fermion's orientation by its isospin.
program weak_twin
  implicit none
  integer, parameter :: dp = selected_real_kind(15, 307)
  integer, parameter :: i8 = selected_int_kind(18)
  integer :: checks, fails
  integer(i8) :: seed
  checks = 0; fails = 0; seed = 20260930_i8
  call block_a(); call block_b(); call block_c(); call block_d();
  call block_e(); call block_f(); call block_g()
  write(*, '(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', checks,
  ', "failures":', fails, '}'
  if (fails > 0) error stop 1
contains
  subroutine check(name, ok)
    character(*), intent(in) :: name
    logical, intent(in) :: ok
    checks = checks + 1
    if (.not. ok) fails = fails + 1
    write(*, '(a,a)') merge(' PASS ', ' FAIL ', ok), name
  end subroutine check

  function rnd() result(r)
    real(dp) :: r
    seed = modulo(16807_i8*seed, 2147483647_i8)
    r = real(seed, dp)/2147483647.0_dp
  end function rnd

  subroutine block_a()
    integer :: l, r, n, neq, nline, t3, q, s, nz
    write(*, '(a)') 'A · parity is the fold on couplings: vector
    couplings on the line, the charged current off it'
    n = 0; neq = 0; nline = 0
    do l = -30, 30
      do r = -30, 30
        n = n + 1
        if (1 + (r - l) == 2 - (1 + (l - r))) .and. r + l == l + r)
          neq = neq + 1
          if ((1 + (l - r) == 1) .eqv. (l == r)) nline = nline + 1
        end do
      end do
      call check('equivariance on 3721 couplings: parity is the fold',
        neq == n)
      call check('a coupling is on the line exactly when it treats
        both hands alike (3721)', nline == n)
      nz = 0
      do t3 = -1, 1
        do q = -3, 3
          do s = 0, 50
            if ((t3 - q*s) - (-q*s)) == t3) nz = nz + 1
          end do
        end do
      end do
      call check('the Z's left-right difference is the weak isospin
        at every charge and mixing weight (1071)', nz == 1071)
    end subroutine block_a
```

```
subroutine block_b()
  integer :: t, nd, nt
  real(dp) :: a(3), b(3), c(3), d1, d2, t1, t2
  write(*, '(a)') 'B · the mirror: every dot product is blind,
  every triple product reverses'
  nd = 0; nt = 0
  do t = 1, 5000
    a = [rnd(), rnd(), rnd()] - 0.5_dp; b = [rnd(), rnd(), rnd()] -
    0.5_dp; c = [rnd(), rnd(), rnd()] - 0.5_dp
    d1 = sum(a*b); d2 = sum((-a)*(-b))
    if (d1 == d2) nd = nd + 1
    t1 = trip(a, b, c); t2 = trip(-a, -b, -c)
    if (t2 == -t1) nt = nt + 1
  end do
  call check('5000 random triples: dot products unchanged by the
  mirror, exactly', nd == 5000)
  call check('5000 random triples: the triple product reversed by
  the mirror, exactly', nt == 5000)
end subroutine block_b

pure function trip(a, b, c) result(t)
  real(dp), intent(in) :: a(3), b(3), c(3)
  real(dp) :: t
  t = a(1)*b(2)*c(3) - b(3)*c(2) - a(2)*b(1)*c(3) - b(3)*c(1) +
  a(3)*b(1)*c(2) - b(2)*c(1))
end function trip

subroutine block_c()
  integer :: nc, np, ncp
  write(*, '(a)') 'C · the charged current: left-handed particles
  and right-handed antiparticles'
  call table(nc, np, ncp)
  call check('C moves the coupling on every state, P on every
  state, CP on none (4 states each)', &
    nc == 4 .and. np == 4 .and. ncp == 0)
end subroutine block_c

subroutine table(nc, np, ncp)
  integer, intent(out) :: nc, np, ncp
  integer :: a, h
  logical :: s0, sc, sp, scp
  nc = 0; np = 0; ncp = 0
  do a = 0, 1
    do h = 0, 1
      s0 = (a == h); sc = ((1 - a) == h); sp = (a == (1 - h)); scp
      = ((1 - a) == (1 - h))
      if (s0 .neqv. sc) nc = nc + 1
      if (s0 .neqv. sp) np = np + 1
      if (s0 .neqv. scp) ncp = ncp + 1
    end do
  end do
end subroutine table

subroutine block_d()
  real(dp) :: s12, s13, s23, c12, c13, c23, d1, jar, worst
  complex(dp) :: v(3,3), e, u(3,3)
  integer :: i, j, n
  integer, parameter :: expect(6) = [0, 0, 1, 3, 6, 10]
  write(*, '(a)') 'D · the mixing matrix: unitary, one phase at
  three generations, J = 3.1e-5'
  s12 = 0.22501_dp; s13 = 0.003732_dp; s23 = 0.04183_dp; d1 =
  1.147_dp
  c12 = sqrt(1.0_dp - s12**2); c13 = sqrt(1.0_dp - s13**2); c23 =
  sqrt(1.0_dp - s23**2)
  e = cmplx(cos(d1), sin(d1), dp)
  v(1,:) = [cmplx(c12*c13, 0.0_dp, dp), cmplx(s12*c13, 0.0_dp,
  dp), s13/e]
```

```

v(2,:) = [-s12*c23 - c12*s23*s13*e, c12*c23 - s12*s23*s13*e,
cmplx(s23*c13, 0.0_dp, dp)]
v(3,:) = [s12*s23 - c12*c23*s13*e, -c12*s23 - s12*c23*s13*e,
cmplx(c23*c13, 0.0_dp, dp)]
u = matmul(v, conjg(transpose(v)))
worst = 0.0_dp
do i = 1, 3
  do j = 1, 3
    worst = max(worst, abs(u(i,j) - merge(1.0_dp, 0.0_dp, i ==
j)))
  end do
end do
jar = aimag(v(1,2)*v(2,3)*conjg(v(1,3))*conjg(v(2,2)))
write(*,'(a,es9.2,a,es11.4)') ' |V V* - 1| <= ', worst, '
Jarlskog invariant J = ', jar
call check('the three-generation mixing matrix is unitary to
1e-15', worst < 1.0e-15_dp)
call check('its one phase gives J = 3.1e-5, the measure of CP
violation in quarks', abs(jar/3.1e-5_dp - 1.0_dp) < 0.03_dp)
n = 0
do i = 1, 6
  if ((i - 1)*(i - 2)/2 == expect(i)) n = n + 1
end do
call check('physical phases (n-1)(n-2)/2 for n = 1..6 are 0, 0,
1, 3, 6, 10: none below three', n == 6)
end subroutine block_d

subroutine block_e()
  real(dp) :: me, mmu, mpi, tree, noha
  write(*,'(a)') 'E · helicity suppression: the pion decays to the
muon, not the electron'
  me = 0.51099895_dp; mmu = 105.6583755_dp; mpi = 139.57039_dp
  noha = ((mpi**2 - me**2)/(mpi**2 - mmu**2))**2
  tree = (me/MMU)**2*noha
  write(*,'(a,es11.4,a,f6.3,a)') ' Gamma(e nu)/Gamma(mu nu) at
tree level = ', tree, ' (phase space alone: ', noha, ')'
  call check('the left-handed coupling suppresses the electron
channel to 1.28e-4, measured 1.2327e-4', &
    abs(tree/1.2327e-4_dp - 1.0_dp) < 0.05_dp .and. noha >
5.0_dp)

```

A · parity is the fold on couplings: vector couplings on the line, the charged current off it
 PASS equivariance on 3721 couplings: parity is the fold
 PASS a coupling is on the line exactly when it treats both hands alike (3721)
 PASS the Z's left-right difference is the weak isospin at every charge and mixing weight (1071)
 B · the mirror: every dot product is blind, every triple product reverses
 PASS 5000 random triples: dot products unchanged by the mirror, exactly
 PASS 5000 random triples: the triple product reversed by the mirror, exactly
 C · the charged current: left-handed particles and right-handed antiparticles
 PASS C moves the coupling on every state, P on every state, CP on none (4 states each)
 D · the mixing matrix: unitary, one phase at three generations, J = 3.1e-5
 |V V* - 1| <= 1.11E-16; Jarlskog invariant J = 3.1170E-05
 PASS the three-generation mixing matrix is unitary to 1e-15
 PASS its one phase gives J = 3.1e-5, the measure of CP violation in quarks

```

end subroutine block_e

subroutine block_f()
  real(dp) :: edm, sm
  write(*,'(a)') 'F · the other forces do not read the
orientation: the neutron's electric dipole'
  edm = 1.8e-26_dp; sm = 1.0e-32_dp
  write(*,'(a,es9.2,a,es9.2,a)') ' neutron EDM < ', edm, ' e cm;
the weak force alone induces about ', sm, ' e cm'
  call check('no time- and parity-odd reading beyond the weak
force is seen: the bound sits 6 orders above it', &
    edm/sm > 1.0e5_dp)
end subroutine block_f

subroutine block_g()
  real(dp) :: sw, anu, ae, au, ad
  write(*,'(a)') 'G · the Z reads each fermion's orientation by
its isospin: A_f = 2 gV gA / (gV^2 + gA^2)'
  sw = 0.23153_dp
  anu = asym(0.5_dp, 0.0_dp, sw); ae = asym(-0.5_dp, -1.0_dp, sw)
  au = asym(0.5_dp, 2.0_dp/3.0_dp, sw); ad = asym(-0.5_dp,
-1.0_dp/3.0_dp, sw)
  write(*,'(a,f7.4,a,f7.4,a,f7.4,a,f7.4)') ' neutrino ', anu, '
charged lepton ', ae, ' up-type ', au, ' down-type ', ad
  call check('the neutrino is purely left-handed to the Z: A = 1
exactly', anu == 1.0_dp)
  call check('lepton, charm and bottom asymmetries 0.147, 0.668,
0.936 against measured 0.1513, 0.670, 0.923', &
    abs(ae/0.1513_dp - 1.0_dp) < 0.035_dp .and.
    abs(au/0.670_dp - 1.0_dp) < 0.035_dp .and. &
    abs(ad/0.923_dp - 1.0_dp) < 0.035_dp)
end subroutine block_g

pure function asym(t3, q, sw) result(a)
  real(dp), intent(in) :: t3, q, sw
  real(dp) :: a, gl, gr, gv, ga
  gl = t3 - q*sw; gr = -q*sw
  gv = gl + gr; ga = gl - gr
  a = 2.0_dp*gv*ga/(gv*gv + ga*ga)
end function asym
end program weak_twin

```

PASS physical phases (n-1)(n-2)/2 for n = 1..6 are 0, 0, 1, 3, 6, 10: none below three
 E · helicity suppression: the pion decays to the muon, not the electron
 Gamma(e nu)/Gamma(mu nu) at tree level = 1.2833E-04 (phase space alone: 5.487)
 PASS the left-handed coupling suppresses the electron channel to 1.28e-4, measured 1.2327e-4
 F · the other forces do not read the orientation: the neutron's electric dipole
 neutron EDM < 1.80E-26 e cm; the weak force alone induces about 1.00E-32 e cm
 PASS no time- and parity-odd reading beyond the weak force is seen: the bound sits 6 orders above it
 G · the Z reads each fermion's orientation by its isospin: A_f = 2 gV gA / (gV^2 + gA^2)
 neutrino 1.0000; charged lepton 0.1470; up-type 0.6675; down-type 0.9355
 PASS the neutrino is purely left-handed to the Z: A = 1 exactly
 PASS lepton, charm and bottom asymmetries 0.147, 0.668, 0.936 against measured 0.1513, 0.670, 0.923
 BATTERY-JSON: {"checks":13,"failures":0}

APPENDIX C: THE SEAT AND THE JUDGMENT

Phys05^I 1.0.5p · RECEIPT · mode quick
 TOOLCHAIN Lean (version 4.19.0, x86_64-unknown-linux-gnu, commit 6caaee842e94, Release) · GNU Fortran (Ubuntu 13.3.0-6ubuntu2-24.04.1) 13.3.0
 MANIFEST sha256 a4108cd26bcbe6ef, the files that ran
 GROUND toolchain: Lean 4.19.0 and gfortran present · manifest: 31 files signed, all verified · proofs: 2 Phys05 Proofs read from II.10, record lines complete · source screen: 15 kernels, 23 named premises, clean · citations: 253 laws named in Parts I and II, 253 resolved · quarantine: 6 retirements string-scanned, 3 held by review, clean
 EXITS ground 0 · registerA RA_TOE_Thesis 0 · registerB Bridge_Final 0 · registerB Universal_Closure 0 · registerB One_Cut 0 · registerB One_Cut_Resolution 0 · registerB One_Cut_Terminal 0 · registerB Only_The_Arrow_Remains 0 · registerB Physical_Closure 0 · registerB Forced_Closure 0 · registerB Closure_Executed 0 · registerB Armed_Seat 0 · registerB Census 0 · registerB Heat_Bridge 0 · registerB The_Loop 0 · registerB RH_Least_Erasure 0 · witness One_Cut_Twin 0 · witness Physical_Closure_Twin 0 · witness Census_Twin 0 · witness Loop_Twin 0 · witness RH_Seal_Twin 0
 REGISTER A {"checks":1123,"failures":0,"mode":"sealed"}
 WITNESS 5 of 5 twins ran, each held to one battery line with zero failures
 CHAIN · D0 83f872725ad9 -> D1 90fc702bbed1 -> D2 492e519daab9 -> D3 6618bec5164b
 ELAPSED ground 1 s · Register A 3 s · Register B 34 s · witness 6 s · total 44 s (not hashed)
 SEAT EARNED · this run

SCREEN forbidden: none | imports: 0 | axioms declared: 0
 laws: 59
 JUDGMENT refused 59 of 59 as proof failures | survived none | parse failures none | planted vacuous law survives its negation: True